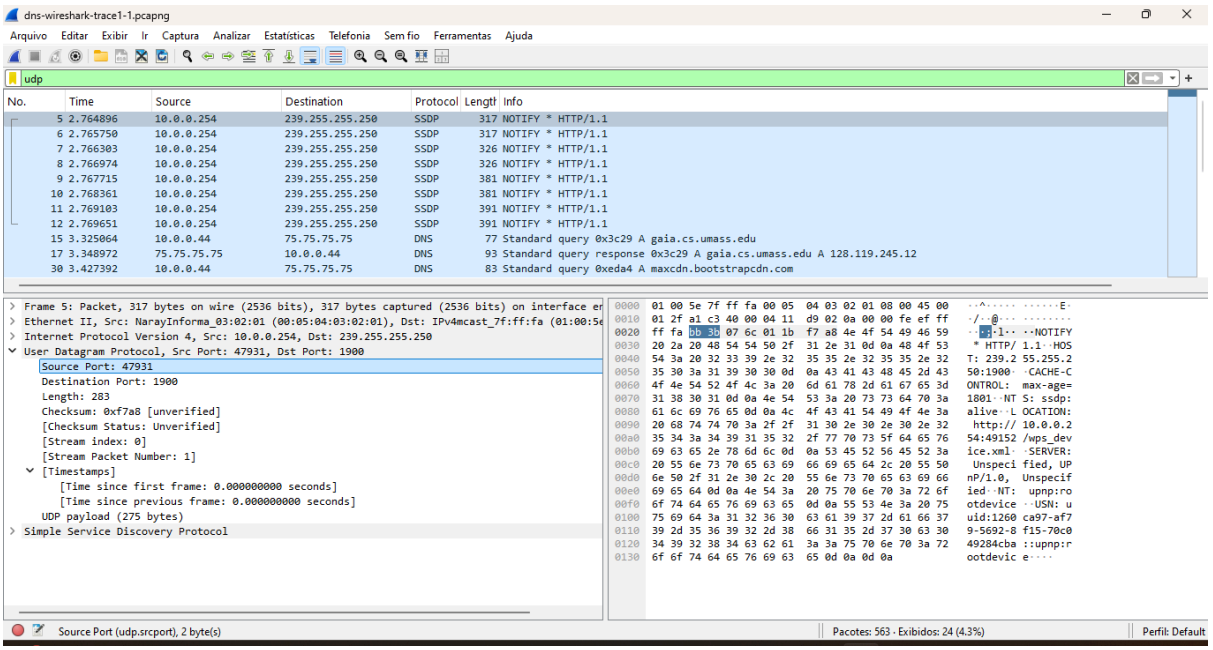
Leonardo H. C. Oliveira - 12311BCC003

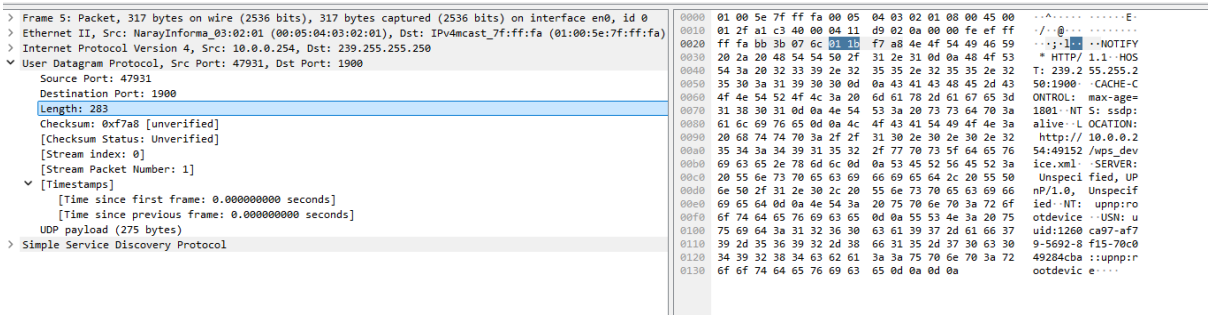
1. Select the first UDP segment in your trace. What is the packet number of this segment in the trace file? What type of application-layer payload or protocol message is being carried in this UDP segment? Look at the details of this packet in Wireshark. How many fields there are in the UDP header? (You shouldn't look in the textbook! Answer these questions directly from what you observe in the packet trace.) What are the names of these fields?

O número do pacote é 5. O protocolo de aplicação é o Simple Service Discovery Protocol. A quantidade de campos é 4. O nome desses campos são: Source Port, Destination Port, Length, Checksum.



2. By consulting the displayed information in Wireshark's packet content field for this packet (or by consulting the textbook), what is the length (in bytes) of each of the UDP header fields?

O comprimento de cada um dos campos do cabeçalho UDP é 2 bytes. Como são 4 campos, a soma do cabeçalho e 8 bytes.



3. The value in the Length field is the length of what? (You can consult the text for this answer). Verify your claim with your captured UDP packet.

O valor no campo Length é o comprimento do datagrama UDP. No caso é 283, sendo a soma do cabeçalho UDP(8 bytes) com o tamanho do payload(275).

```
User Datagram Protocol, Src Port: 47931, Dst Port: 1900
    Source Port: 47931
    Destination Port: 1900
    Length: 283
    Checksum: 0xf7a8 [unverified]
    [Checksum Status: Unverified]
    [Stream index: 0]
    [Stream Packet Number: 1]
  [Timestamps]
      [Time since first frame: 0.000000000 seconds]
      [Time since previous frame: 0.000000000 seconds]
    UDP payload (275 bytes)
```

4. What is the maximum number of bytes that can be included in a UDP payload? (Hint: the answer to this question can be determined by your answer to 2. above)

Pelo campo length do cabeçalho UDP ter 16 bits, o tamanho total do datagrama é 65535 bytes. Mas como o cabeçalho possui tamanho de 8 bytes, é necessário subtrair esse número do resultado anterior. Fazendo isso, chega na resposta final: 65527 bytes.

5. What is the largest possible source port number? (Hint: see the hint in 4.)

Como source port possui 16 bits, o maior valor que ele pode assumir é 65535 bytes.

6. What is the protocol number for UDP? Give your answer in decimal notation. To answer this question, you'll need to look into the Protocol field of the IP datagram containing this UDP segment (see Figure 4.13 in the text, and the discussion of IP header fields).

O número do protocolo UDP é 17.

```
Internet Protocol Version 4, Src: 10.0.0.254, Dst: 239.255.255.250
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
  Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
      0000 00.. = Differentiated Services Codepoint: Default (0)
      .... ..00 = Explicit Congestion Notification: Not ECN-Capable Transport (0)
    Total Length: 303
    Identification: 0xa1c3 (41411)
  010. .... = Flags: 0x2, Don't fragment
      0... .... = Reserved bit: Not set
      .1.. .... = Don't fragment: Set
      ..0. .... = More fragments: Not set
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 4
    Protocol: UDP (17)
    Header Checksum: 0xd902 [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 10.0.0.254
    Destination Address: 239.255.255.250
    [Stream index: 0]
```

7. Examine the pair of UDP packets in which your host sends the first UDP packet and the second UDP packet is a reply to this first UDP packet. (Hint: for a second packet to be sent in response to a first packet, the sender of the first packet should be the destination of the second packet). What is the packet number of the first of these two UDP segments in the trace file? What is the value in the source port field in this UDP segment? What is the value in the destination port field in this UDP segment? What is the packet number of the second of these two UDP segments in the trace file? What is the value in the source port field in this second UDP segment? What is the value in the destination port field in this second UDP segment? Describe the relationship between the port numbers in the two packets

O número do primeiro é 15. O valor do Source Port é 58350. O valor do Destination Port é 53. O número de pacote do segundo é 17. O valor do Source Port é 53. O valor do Destination Port é 58350.

```
15 3.325064    10.0.0.44       75.75.75.75     DNS      77 Standard query 0x3c29 A gaia.cs.umass.edu
17 3.348972    75.75.75.75     10.0.0.44       DNS      93 Standard query response 0x3c29 A gaia.cs.umass.edu A 128.119.245.12
```

```
Source Port: 58350
Destination Port: 53
```

```
User Datagram Protocol, Src Port: 53, Dst Port: 58350
    Source Port: 53
    Destination Port: 58350
```